

**PLAYER PERFORMANCE PREDICTION USING
RIDGE AND LASSO REGRESSION**

project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY
in
A.Y.: 2026-27BY**

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DEPARTMENT OF CSE

Name of Title: Player Performance Prediction Using Ridge and Lasso Regression

1. ABSTRACT:

The rapid growth of data analytics has significantly transformed the field of sports by enabling teams to make informed decisions based on player statistics and performance trends. Predicting a player's future performance has become an important aspect of sports analytics, as it helps coaches, selectors, and team management evaluate players objectively and develop effective strategies. Traditional methods of player evaluation often rely on experience and subjective judgment, which may not always provide accurate or consistent results. This project aims to address this challenge by developing a Machine Learning model that predicts player performance using historical statistical data. The proposed system employs two supervised Machine Learning algorithms, **Ridge Regression** and **Lasso Regression**, to analyze the relationship between a player's previous performance and future outcomes. These algorithms are chosen because they are capable of improving prediction accuracy while reducing the problem of overfitting. Ridge Regression minimizes the impact of multicollinearity by shrinking regression coefficients, whereas Lasso Regression performs feature selection by assigning zero coefficients to less significant features, thereby simplifying the model. A comparative analysis of these algorithms helps identify the model that provides more reliable and accurate predictions. The implementation begins with collecting a dataset containing player statistics such as matches played, runs scored, batting average, strike rate, centuries, and other relevant performance indicators. The dataset undergoes preprocessing steps including handling missing values, removing duplicate records, feature selection, and feature scaling to improve data quality. Exploratory Data Analysis (EDA) is then performed to understand the distribution of the data, identify patterns, and examine relationships among different variables before training the models. The trained models are evaluated using performance metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared (R^2) score. The project compares the predictive performance of Ridge and Lasso Regression to determine the most suitable model. The proposed system demonstrates how Machine Learning can provide accurate, data driven insights for player performance prediction, assist in team selection and strategic planning, and contribute to the growing field of sports analytics.

2. INTRODUCTION:

The advancement of Machine Learning has revolutionized the way data is analyzed and utilized across various industries, including healthcare, finance, education, and sports. In the field of sports analytics, large volumes of player statistics are generated from every match, providing valuable information that can be used to evaluate performance and make strategic decisions. Analyzing this data manually is often challenging, time-consuming, and prone to subjective judgment. Machine Learning provides an efficient solution by identifying hidden patterns within historical data and generating accurate predictions.

Player performance prediction has become an important application of Machine Learning in modern sports. Accurate performance forecasting helps coaches, selectors, and team management assess players objectively, identify promising talent, formulate match strategies, and improve overall team performance. Instead of relying solely on intuition or past experience, data-driven prediction models enable more reliable and consistent decision-making. Regression techniques are widely used when the objective is to predict a continuous numerical value. In this project, Ridge Regression and Lasso Regression are employed to predict player performance using historical statistics such as matches played, runs scored, batting average, strike rate, centuries, and other relevant performance indicators. Both algorithms are extensions of Linear Regression that incorporate regularization techniques to reduce overfitting and improve the model's ability to generalize to unseen data. Ridge Regression minimizes the impact of multicollinearity by shrinking coefficient values, whereas Lasso Regression performs feature selection by reducing the coefficients of less important features to zero, resulting in a simpler and more interpretable model.

The project follows a systematic Machine Learning workflow that includes data collection, preprocessing, exploratory data analysis, feature selection, feature scaling, model training, prediction, and performance evaluation. The trained models are compared using evaluation metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the R-squared (R^2) score to determine the most suitable model for player performance prediction.

The primary objective of this project is to develop an accurate and reliable prediction system that can estimate player performance based on historical data while comparing the effectiveness of Ridge Regression and Lasso Regression. The project demonstrates how Machine Learning can transform raw sports data into meaningful insights, enabling data-driven decision-making and contributing to the growing field of sports analytics.

3. SOFTWARE REQUIREMENTS

The proposed system is developed using Python 3.x in Google Colaboratory (Google Colab), which provides a cloudbased environment for writing and executing Python programs without requiring local software installation.

The project utilizes the following Python libraries:

- Pandas – For data loading and preprocessing.
- NumPy – For numerical computations.
- Matplotlib – For data visualization.
- Scikit-learn (sklearn) – For implementing Ridge Regression, Lasso Regression, feature scaling, model training, and performance evaluation.

A publicly available CSV dataset containing player performance statistics is used for training and testing the models. The system can be accessed through any modern web browser with an active internet connection.


Signature of the Guide


Course Instructor